

Euclidean Distance in Python

In the beginning of this project, I thought of it much like the calculator project we had. Instead of defining an operator and having two numbers, for this I would need to define the points that I would need and then apply them to the “operation” to get the Euclidean distance. Making the equation for the Euclidean distance ended up being the easiest in this project:

```
dist2 = ((abs)(p1x - p2x)**2 + (abs)(p1y - p2y)**2)**(1/2)
```

For the Euclidean distance I subtracted my first and second x values and squared them, I did the same with the y values and added them together. For this formula I originally imported math to square root the value. I later added (abs) to get the absolute value of the x and y values as well as the exponent of $\frac{1}{2}$. After this I defined what my point values would be named by:

```
p1x = float(x)
p1y = float(y)
p2x = float(c)
p2y = float(v)
```

Finally I made a way to input the desired point values and printing the result.

```
x = input("first x: ")
y = input("first y: ")
c = input("second x: ")
v = input("second y: ")
```

```
print ("distance:", dist2)
```

The difficulties I had with this project were first I had trouble pulling a specific value out of a list using [], when I had point values arranged in a list as [x,y] I was able to pull them out to do basic adding or subtracting. When I tried to put them into the equation, I made for the Euclidean distance, it would basically output an error that it couldn't subtract a string and a string. The second problem I couldn't figure out was how to square root without importing the library math.

My solutions to the challenges, the easiest one to figure out was the square root problem. I was doing math homework a little after making this program and realized raising something to $\frac{1}{2}$ is the exact same as square root. I went back and changed my equation to include raising the whole thing by $\frac{1}{2}$ and that fixed my issue of not including the math library into the equation. The second problem took me a little longer to figure out. I tried making an array at first and seeing if that changed anything, it came back with the same error. Next, I looked into making my list a tuple but that didn't work either, I don't know if I was not using it correctly or that I wasn't able

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to use it in this fashion in the first place. Finally, I decided to make each point separate and not in a list of multiple things, this didn't change how I was inputting values so all I had to change was defining the points separately and changing what was going into the Euclidean calculation. The most recent change that I made was to include (abs) to get the absolute value after combining the x and y values. I left it out at first because I figured that if I was raising these values to the power of 2 I wouldn't get any negatives from this. I later added (abs) just to get practice applying this but my original equation didn't include it.